Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: A & C Vieira Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
31503 County Road 26	Woodland	Yolo	95695

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/27/1999

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

0050-0011-0003-0000

B. OPERATORS

Vieira, Angelo

Operator name: Vieira, Angelo

Telephone no.: Landline: ; Cellular: (209) 678-2679

Mailing Address Number and Street	City	State	Zip Code
8345 Olive School LN	Winters	CA	95694

This operator is responsible for paying permit fees.

C. OWNERS

Kasbergen, Jack

Legal owner name: Kasbergen, Jack

Telephone no.: Landline: (530) 400-7019; Cellular:

Mailing Address Number and Street	City	State	Zip Code
31870 County Road 27	Woodland	CA	95695

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1,339	160	400	0	500	300
Maximum number	1,339	160	400	0	500	300
Average number	1,339	160	400	0	500	300
Avg live weight (lbs)	1,100	1,200	900	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 42,399.83 *tons per reporting period*

Total nitrogen from manure: 544,786.10 *lbs per reporting period*

After ammonia losses (30% loss applied): 381,350.27 *lbs per reporting period*

Total phosphorus from manure: 86,927.97 *lbs per reporting period*

Total potassium from manure: 251,152.52 *lbs per reporting period*

Total salt from manure: 667,260.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,873,000 *gallons*

Total nitrogen generated: 43,374.01 *lbs*

Total phosphorus generated: 3,264.08 *lbs*

Total potassium generated: 74,005.38 *lbs*

Total salt generated: 369,536.79 *lbs*

	5,873,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	5,873,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Yolo County Flood Control and Water Conservation District	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/06/2023	Solid commercial fertilizer *Fertilizer*	233.86 *ton*	Dry-weight	1.0	46.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	212,999.69	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	212,999.69	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy Field	270	270	2	both	X040-X009-X001-XXXX X050-X011-X003-XXXX
East Block	320	320	2	manure	X040-X009-X002-XXXX X040-X009-X004-XXXX
Jack's Horse Field	120	120	2	manure	X050-X011-X003-XXXX
Totals for areas that were used for application	710	710	6		
Totals for areas that were not used for application					
Land application area totals	710	710	6		

B. CROPS AND HARVESTS

Dairy Field

Field name: Dairy Field

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 270 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	4,281.00 *ton*	Dry-weight		38.8	10,800.00	2,370.00	19,200.00		9.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	15.86	209.60	46.00	372.62	1,849.51

Dairy Field

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 270 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	7,885.00 *ton*	Dry-weight		66.5	11,700.00	3,130.00	12,500.00		4.81

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	29.20	228.93	61.24	244.58	941.15

East Block

Field name: East Block

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 320 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	4,854.00 *ton*	Dry-weight		28.8	11,600.00	2,930.00	23,000.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	15.17	250.56	63.29	496.81	2,527.24

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 320 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	9,653.00 *ton*	Dry-weight		66.2	12,500.00	3,300.00	13,800.00		4.91

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	30.17	254.90	67.29	281.41	1,001.25

Jack's Horse Field

Field name: Jack's Horse Field

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 120 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	1,880.00 *ton*	Dry-weight		55.9	11,300.00	2,170.00	16,500.00		8.51

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	15.67	156.14	29.99	228.00	1,175.91

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 120 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	3,596.00 *ton*	Dry-weight		66.5	11,500.00	1,990.00	20,200.00		6.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	29.97	230.89	39.95	405.57	1,242.81

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy Field - 11/10/2022: Oats, silage-soft dough

Field name: Dairy Field

Crop: Oats, silage-soft dough — Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	79.36	21.28	88.61	1,003.53	957.00 *ton*
Application event totals		79.36	21.28	88.61	1,003.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	160.64	12.09	274.09	1,368.65	5,873,000.00 *gal*
Application event totals		160.64	12.09	274.09	1,368.65	

Dairy Field - 06/04/2023: Corn, silage

Field name: Dairy Field

Crop: Corn, silage — Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fertilizer	Solid commercial fertilizer	300.00	0.00	0.00	0.00	
Application event totals		300.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.14	0.00	0.00	21.37	34,565,000.00 *gal*
Application event totals		2.14	0.00	0.00	21.37	

Dairy Field - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.15	0.00	0.00	21.53	34,834,000.00 *gal*
Application event totals		2.15	0.00	0.00	21.53	

07/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.10	0.00	0.00	20.95	33,897,000.00 *gal*
Application event totals		2.10	0.00	0.00	20.95	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.05	0.00	0.00	20.49	33,152,000.00 *gal*
Application event totals		2.05	0.00	0.00	20.49	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.04	0.00	0.00	20.40	33,005,000.00 *gal*
Application event totals		2.04	0.00	0.00	20.40	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	1.98	0.00	0.00	19.84	32,099,000.00 *gal*
Application event totals		1.98	0.00	0.00	19.84	

East Block - 11/08/2022: Oats, silage-soft dough

East Block - 11/08/2022: Oats, silage-soft dough

Field name: East Block

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	271.37	88.14	292.74	3,279.96	3,957.00 *ton*
Application event totals		271.37	88.14	292.74	3,279.96	

East Block - 06/09/2023: Corn, silage

Field name: East Block

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.19	0.00	0.00	21.91	42,003,000.00 *gal*
Application event totals		2.19	0.00	0.00	21.91	

06/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fertilizer	Solid commercial fertilizer	300.00	0.00	0.00	0.00	
Application event totals		300.00	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.14	0.00	0.00	21.43	41,090,000.00 *gal*
Application event totals		2.14	0.00	0.00	21.43	

East Block - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.13	0.00	0.00	21.31	40,867,000.00 *gal*
Application event totals		2.13	0.00	0.00	21.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.12	0.00	0.00	21.22	40,691,000.00 *gal*
Application event totals		2.12	0.00	0.00	21.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.09	0.00	0.00	20.86	39,997,000.00 *gal*
Application event totals		2.09	0.00	0.00	20.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.01	0.00	0.00	20.13	38,591,000.00 *gal*
Application event totals		2.01	0.00	0.00	20.13	

Jack's Horse Field - 11/12/2022: Oats, silage-soft dough

Field name: Jack's Horse Field

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Jack's Horse Field - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	195.91	52.53	218.75	2,477.37	1,050.00 *ton*
Application event totals		195.91	52.53	218.75	2,477.37	

Jack's Horse Field - 06/12/2023: Corn, silage

Field name: Jack's Horse Field

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fertilizer	Solid commercial fertilizer	300.00	0.00	0.00	0.00	
Application event totals		300.00	0.00	0.00	0.00	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.23	0.00	0.00	22.33	16,057,000.00 *gal*
Application event totals		2.23	0.00	0.00	22.33	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.22	0.00	0.00	22.25	15,997,000.00 *gal*
Application event totals		2.22	0.00	0.00	22.25	

Jack's Horse Field - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.21	0.00	0.00	22.13	15,911,000.00 *gal*
Application event totals		2.21	0.00	0.00	22.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.21	0.00	0.00	22.14	15,918,000.00 *gal*
Application event totals		2.21	0.00	0.00	22.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.10	0.00	0.00	21.00	15,101,000.00 *gal*
Application event totals		2.10	0.00	0.00	21.00	

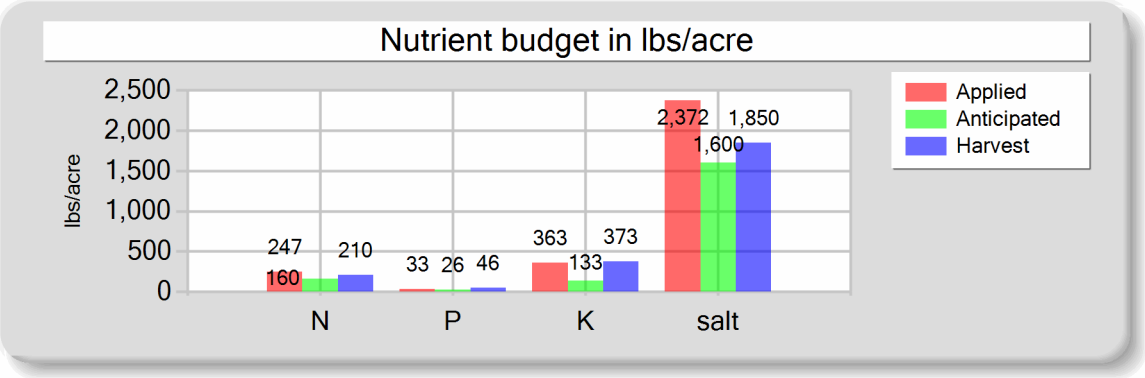
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Yolo County Flood Control and Water Conservation District	Surface water	2.22	0.00	0.00	22.25	15,997,000.00 *gal*
Application event totals		2.22	0.00	0.00	22.25	

B. NUTRIENT BUDGET

Dairy Field - 11/10/2022: Oats, silage-soft dough

Field name: Dairy Field Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79.36	21.28	88.61	1,003.53
Process wastewater	160.64	12.09	274.09	1,368.65
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.00	33.37	362.70	2,372.19
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	209.60	46.00	372.62	1,849.51
Nutrient balance	37.40	-12.63	-9.91	522.68
Applied to removed ratio	1.18	0.73	0.97	1.28

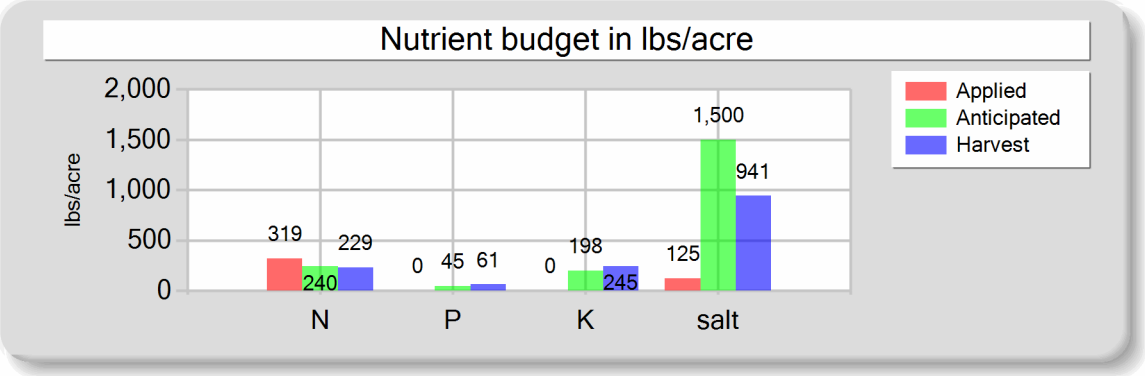
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
5,873,000.00	*gallons*
216.28	*acre-inches*
0.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy Field - 06/04/2023: Corn, silage

Field name: Dairy Field | Crop: Corn, silage | Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	300.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	12.46	0.00	0.00	124.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.46	0.00	0.00	124.59
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	228.93	61.24	244.58	941.15
Nutrient balance	90.53	-61.24	-244.58	-816.56
Applied to removed ratio	1.40	0.00	0.00	0.13

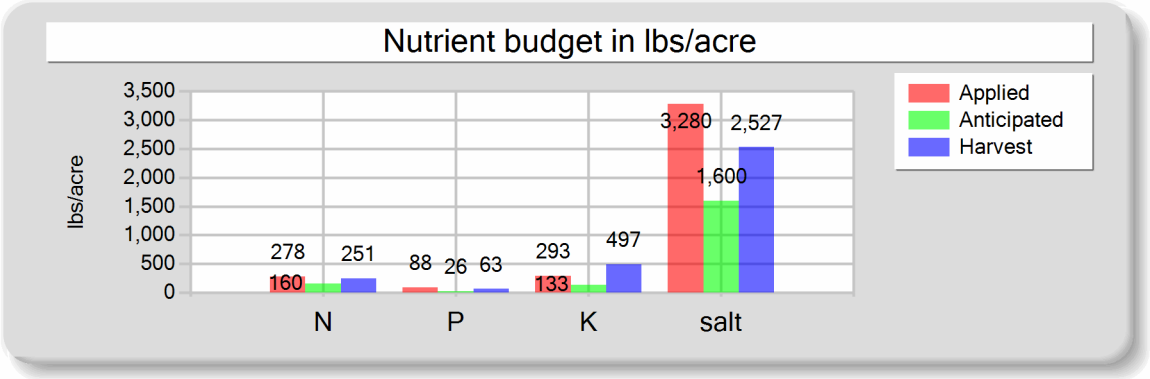
Fresh water applied	
201,552,000.00	*gallons*
7,422.47	*acre-inches*
27.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

East Block - 11/08/2022: Oats, silage-soft dough

Field name: East Block Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	271.37	88.14	292.74	3,279.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.37	88.14	292.74	3,279.96
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	250.56	63.29	496.81	2,527.24
Nutrient balance	27.81	24.85	-204.07	752.72
Applied to removed ratio	1.11	1.39	0.59	1.30

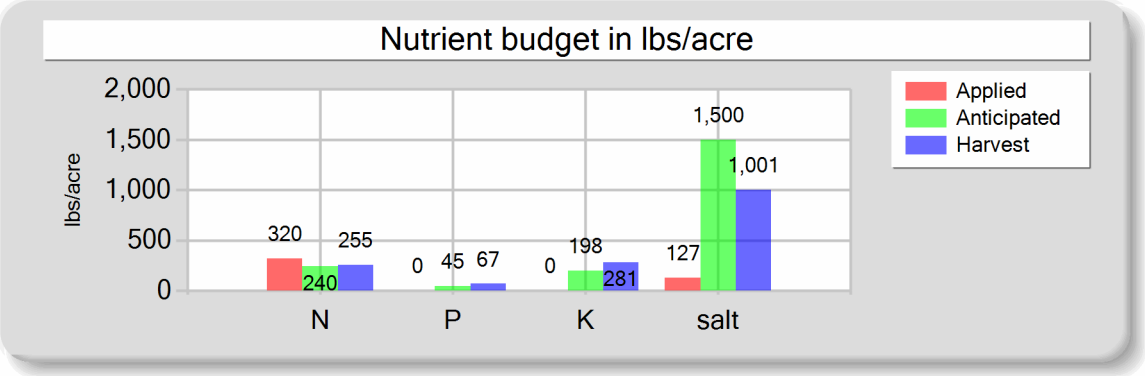
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

East Block - 06/09/2023: Corn, silage

Field name: East Block | Crop: Corn, silage | Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	300.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	12.69	0.00	0.00	126.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.69	0.00	0.00	126.86
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	254.90	67.29	281.41	1,001.25
Nutrient balance	64.79	-67.29	-281.41	-874.38
Applied to removed ratio	1.25	0.00	0.00	0.13

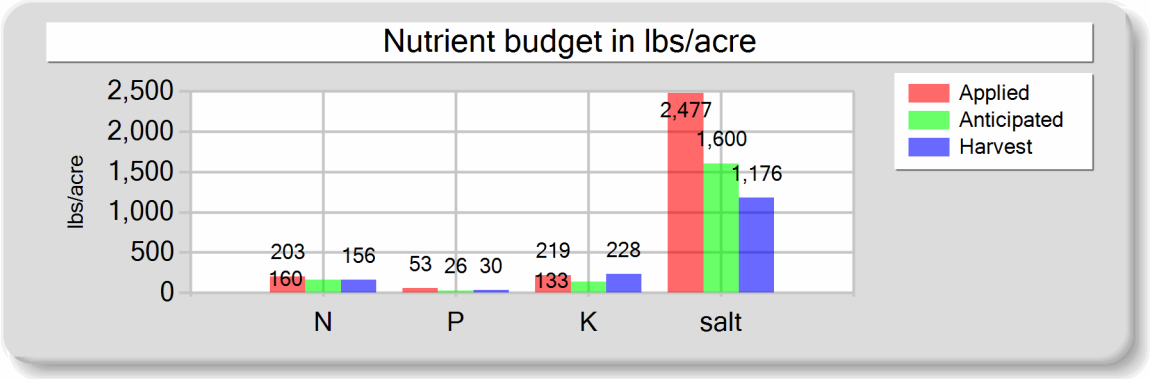
Fresh water applied	
243,239,000.00	*gallons*
8,957.67	*acre-inches*
27.99	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jack's Horse Field - 11/12/2022: Oats, silage-soft dough

Field name: Jack's Horse Field Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	195.91	52.53	218.75	2,477.37
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.91	52.53	218.75	2,477.37
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	156.14	29.99	228.00	1,175.91
Nutrient balance	46.76	22.55	-9.25	1,301.46
Applied to removed ratio	1.30	1.75	0.96	2.11

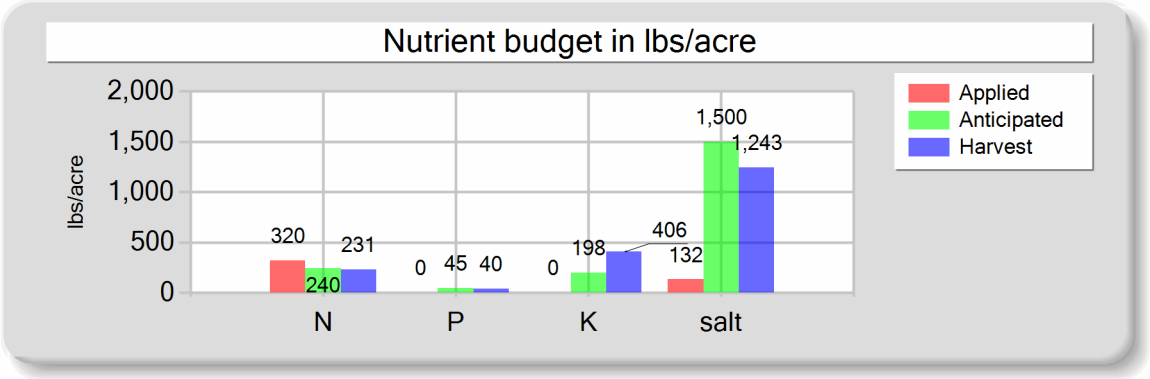
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Jack's Horse Field - 06/12/2023: Corn, silage

Field name: Jack's Horse Field Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	300.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.21	0.00	0.00	132.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.21	0.00	0.00	132.10
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	230.89	39.95	405.57	1,242.81
Nutrient balance	89.32	-39.95	-405.57	-1,110.70
Applied to removed ratio	1.39	0.00	0.00	0.11

Fresh water applied	
94,981,000.00	*gallons*
3,497.83	*acre-inches*
29.15	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

W4005001A

Sample and source description: W4005001A

Sample date: 09/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	5,980.00	24,900.00							28.20
DL	100.00	100.00	100.00							0.10

W4005001B

Sample and source description: W4005001B

Sample date: 10/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,400.00	8,250.00	27,400.00							30.70
DL	100.00	100.00	100.00							0.10

B. PROCESS WASTEWATER ANALYSES

T4008001B

Sample and source description: T4008001B

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	885.00	465.00	0.00	0.00	66.60	1,510.00								10,000.00	7,540
DL	0.10	0.10	0.10	0.10	0.10	0.10								20.00	12

C. FRESH WATER ANALYSES

Yolo County Flood Control and Water Conservation District

District info

Sample description: District info

Sample date: 07/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.00	0.00	2.00								30.00	20
DL	0.10	0.10	0.10								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Dairy Field - 11/10/2022: Oats, silage-soft dough

W4005001C

Sample and source description: W4005001C

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	2,370.00	19,200.00		9.53
DL	100.00	100.00	100.00		0.10

Dairy Field - 06/04/2023: Corn, silage

Dairy Field - 06/04/2023: Corn, silage

W4005001F

Sample and source description: W4005001F

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	3,130.00	12,500.00		4.81
DL	100.00	100.00	100.00		0.10

East Block - 11/08/2022: Oats, silage-soft dough

W4005001D

Sample and source description: W4005001D

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 28.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,930.00	23,000.00		11.70
DL	100.00	100.00	100.00		0.10

East Block - 06/09/2023: Corn, silage

W4005001G

Sample and source description: W4005001G

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	3,300.00	13,800.00		4.91
DL	100.00	100.00	100.00		0.10

Jack's Horse Field - 11/12/2022: Oats, silage-soft dough

W4005001E

Sample and source description: W4005001E

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,170.00	16,500.00		8.51
DL	100.00	100.00	100.00		0.10

Jack's Horse Field - 06/12/2023: Corn, silage

W4005001H

Sample and source description: W4005001H

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	1,990.00	20,200.00		6.19
DL	100.00	100.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

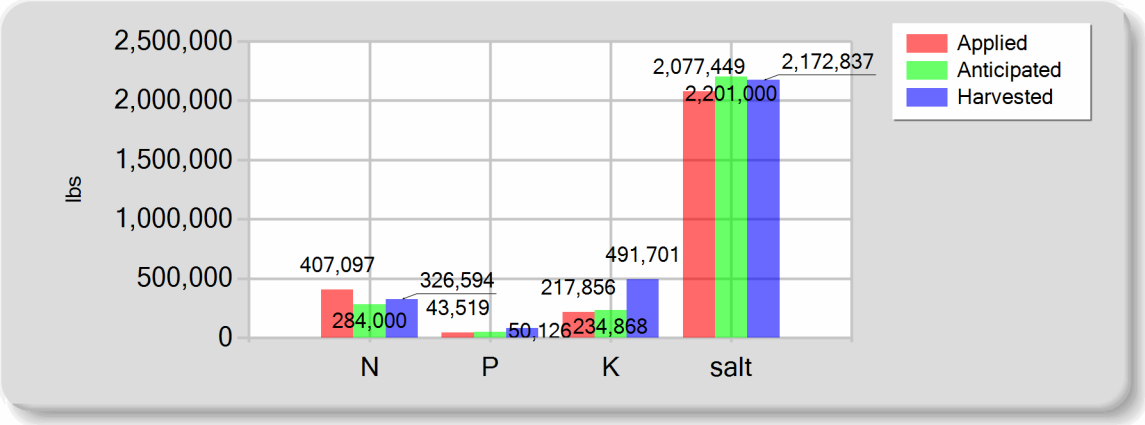
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

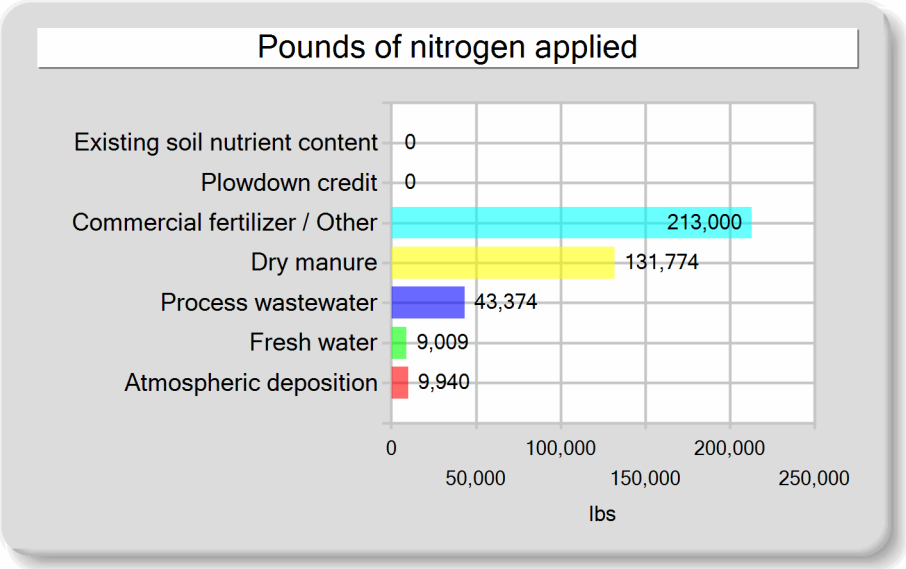
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

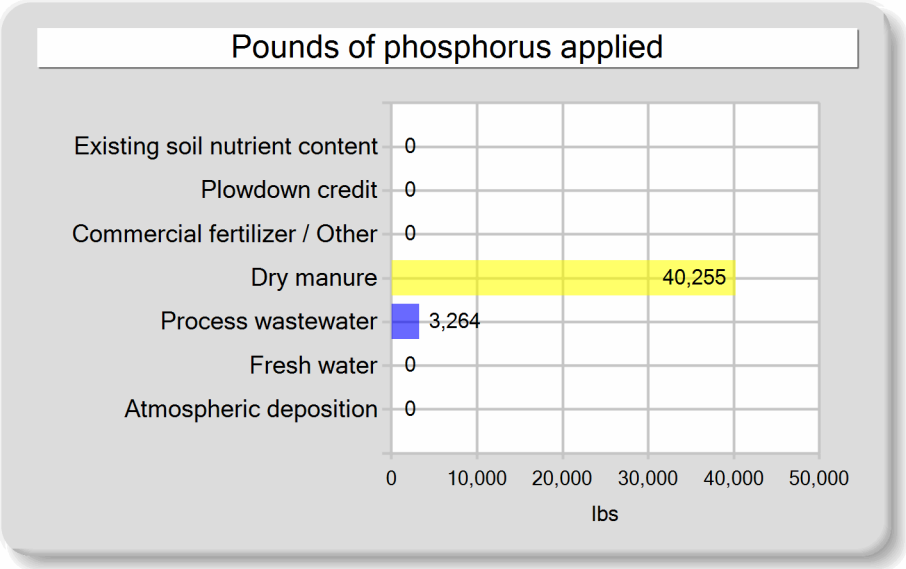
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	213,000.00	0.00	0.00	0.00
Dry manure	131,773.86	40,255.36	143,850.63	1,617,824.23
Process wastewater	43,374.01	3,264.08	74,005.38	369,536.79
Fresh water	9,008.79	0.00	0.00	90,087.95
Atmospheric deposition	9,940.00	0.00	0.00	0.00
Total nutrients applied	407,096.67	43,519.44	217,856.01	2,077,448.97
Anticipated crop nutrient removal	284,000.00	50,126.00	234,868.00	2,201,000.00
Actual crop nutrient removal	326,594.46	79,133.41	491,700.82	2,172,836.63
Nutrient balance	80,502.21	-35,613.97	-273,844.81	-95,387.66
Applied to removed ratio	1.25	0.55	0.44	0.96

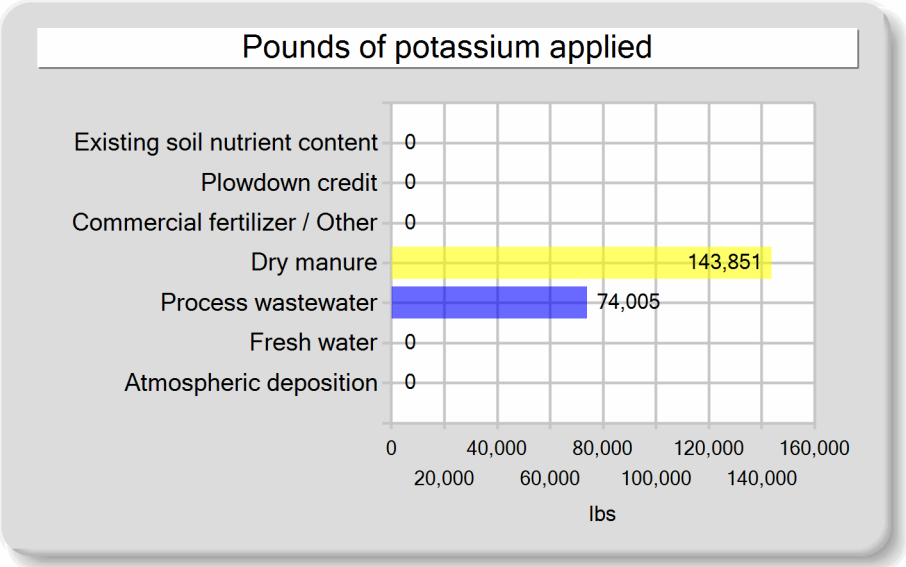
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

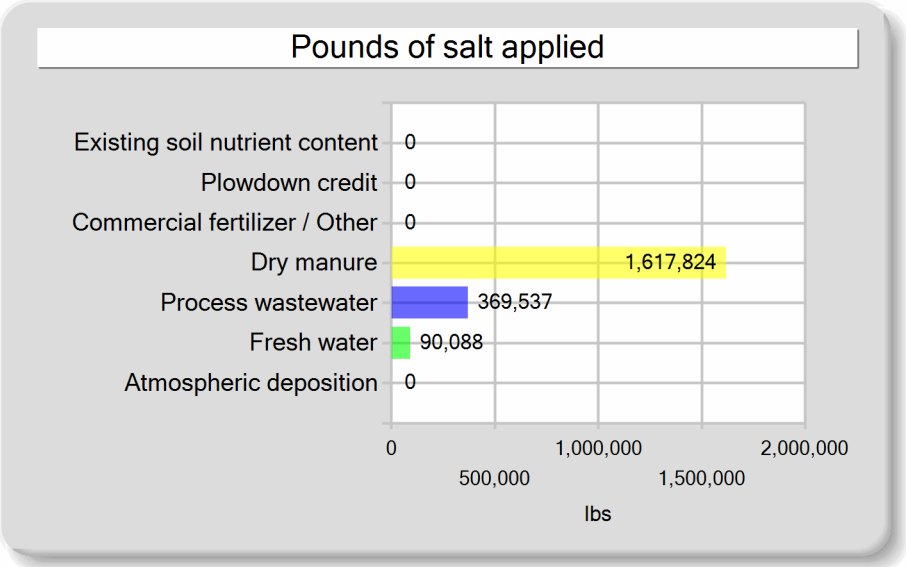


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

	Angelo Vieira
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jack Kasbergen	Angelo Vieira
PRINT OR TYPE NAME	PRINT OR TYPE NAME
	6-29-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
A & C Vieira Dairy
8345 Olive School Ln.
Winters, CA 95694

Grower

Consultant:
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 1/5/24 10:02 AM
SUBMITTED BY: Paul
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Clear Water

NOTES: Temperature above qualifying at 18.9 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well	T4008001A	Electrical Conductivity (EC)	EPA 120.1 and	0.842	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	539	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)	0.619
Electrical Conductivity (EC)	94.9			2.69
Nitrate Nitrogen (NO3-N)	103	102 (T4003010A)	104 (T4003010A)	2.24

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: A&C Vieira Dairy
Mailing address: 8345 Olive School Ln
(not facility address) Winters Ca
Phone#: (209) 678-2679

Authorized Copy Released To: Paul Sousa (209) 556-2490
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

GROUND WATER
, NH3-N*
S
1 of H1 & H6
, NH3-N, Ca, Mg, Na, Bicarb
Cl, TDS

ASTE WATER
, TKN, TP, K, TDS
, NH4-N, TKN, TP, K, TDS
Bicarb, Carb, SO4, Cl
1 of H3 &H5
1 of H4 &H5

, TP, NH4-N*, TDS
Measurement
e if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

ted	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
	12:00pm	well		18.9		✓				
	12:00pm	Lagoon		15.2			✓			
	~~12:00pm~~	~~Solid manure #1~~						✓		ash
	~~12:00pm~~	~~Solid manure #2~~						✓		ash
	~~12:00pm~~	~~oats dairy block~~							✓	
	~~12:00pm~~	~~oats oldman block~~							✓	
	~~12:00pm~~	~~oats Jacks block~~							✓	
	~~1pm~~	~~Corn dairy block~~							✓	
	~~1pm~~	~~Corn Oldman block~~							✓	
	~~1pm~~	~~Corn Jacks block~~							✓	

Analysis

CHAIN OF CUSTODY RECORDING

Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER (LABORATORY USE ONLY)
	A&C Vieira		1-5-24 10:02am		
	DENELE ANALYTICAL, INC.	1-5-24 10:02a			T4000 9021